Heart Disease Predictor Using Machine Learning Project Report

---

Abstract

This project focuses on predicting heart disease using supervised machine learning classification models. Using a structured dataset containing patient health parameters, we trained and compared Logistic Regression, Decision Tree, Random Forest, and a Neural Network classifier. Model performance was evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The Random Forest and Neural Network models demonstrated superior performance. Feature importance analysis provided insights into key risk indicators.

Problem Statement

Heart disease is one of the leading causes of death globally. Early prediction enables effective diagnosis and timely medical intervention. The objective of this project is to develop a machine learning model that predicts whether a patient has heart disease based on clinical parameters such as age, cholesterol levels, chest pain type, blood pressure, ECG results, and more.

Objectives

1. Build supervised ML models: Logistic Regression, Decision Tree, Random Forest, and Neural Network.
2. Evaluate models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
3. Compare performance to identify the best model.
4. Analyze feature importance and identify major risk contributing variables.

Dataset Description

The dataset contains medical parameters commonly used in heart disease diagnosis.
Key features include: - age: Age in years
- sex: Gender (0 = female, 1 = male)
- cp: Chest pain type
- trestbps: Resting blood pressure
- chol: Serum cholesterol

- thalach: Maximum heart rate achieved
- exang: Exercise-induced angina
- oldpeak: ST depression
- ca, thal: Additional clinical indicators

The target variable: 0 = No heart disease, 1 = Heart disease present.

Methodology

1. Data Cleaning: Handled missing values, encoded categorical variables, and scaled features.
2. Exploratory Data Analysis: Visualized correlations and feature distributions.
3. Model Training: Implemented Logistic Regression, Decision Tree, Random Forest, and Neural Network.
4. Evaluation: Computed metrics such as accuracy, recall, precision, F1-Score, and ROCAUC.
5. Feature Importance: Analyzed model specific weights and importance scores.

Model Performance Summary

Logistic Regression performed moderately well, offering interpretability. Decision Tree gave reasonable results but tended to overfit. Random Forest achieved the best overall performance with strong ROC-AUC and balanced precision recall scores. The Neural Network model also performed competitively after feature scaling and hyperparameter tuning.

Feature Importance

Tree based models identified chest pain type (cp), ST depression (oldpeak), maximum heart rate (thalach), and the number of major vessels (ca) as strong predictors. Logistic Regression coefficients also supported these findings.

Conclusion

The project successfully implemented multiple machine learning classification models for predicting heart disease. Random Forest emerged as the best performing model. The analysis highlights clinically relevant features associated with elevated heart disease risk. Future work may include deploying the model as a web app, expanding the dataset, and experimenting with advanced models such as XGBoost.

GitHub Repository

The project code, dataset, model files, and notebooks are available in the GitHub repository (https://github.com/Mayuur-web/Heart-Disease-Predictor).